

Replication Project

Can Warm Behaviour Mitigate the Negative Effect of Unfavourable Governmental Decisions on Citizens' Trust?

Hansen, F. G. (2023). *Journal of Experimental Political Science*, 10(1), 62–75.

Theoretical Background

Context

- Citizen trust is fundamental in successful governance
- Trust is often based on government performance and whether decisions fit individual preferences
- There is some evidence that direct interactions with government staff has an impact on trust in the government
- From social psychology, we understand the importance of interpersonal impressions when people make evaluative judgements

When citizens interact with government employees, what is the impact of the perceived traits of this employee and does this impact broader government trust?

Hypotheses

H1: Citizens have lower trust in government the more unfavourable a decision is to the individual.

H2: The effect of decision favourability on citizens' trust in government depends on whether bureaucrats show warmth in the decision-making process.

The negative effect of unfavourable decisions is weaker when bureaucrats show high warmth compared to showing low warmth.

Methodology

Experimental Groups

		Government Caseworker	
		Warmth = 0	Warmth = 1
Result of Nursing Home Application	Favourability = 0	Group 4 Low Warmth + Unfavourable Decision	Group 1 High Warmth + Unfavourable Decision
	Favourability = 1	Group 3 Low Warmth + Favourable Decision	Group 2 High Warmth + Favourable Decision

2x2 survey experiment on representative sample of Danish Citizens ($n = 1613$)

Method

- Participants read one of four vignettes in which they imagine helping a close elderly relative apply for a place in a nursing home
- Two features are manipulated in the vignettes: the description of the caseworker (high or low warmth) and decision on the application (accepted versus rejected).
- Participants answered a series of demographic questions as well as rating the caseworker and the decision on a number of traits
- Trust was then rated on a 0-10 scale depending on how much they trusted (a) the bureaucrat and (b) the municipality

“How much do you personally trust the [municipality/the caseworker] to make the right decisions?”



Data

Favourability of
decision outcome

Warmth of
Bureaucrat

Trust in
Bureaucrat

Trust in
Municipality

caseid	treatment_groups		of_treatment	warmth_treatment	trust_b	trust_m	gender
1818592968	3	1		0	0.5	0.0	0
1823485372	1	0		1	0.3	0.3	1
1823435322	1	0		1	0.3	0.3	1
1823435332	4	0		0	0.2	0.2	1
1823492020	3	1		0	NA	NA	0
1821707542	1	0		1	0.6	0.5	0
1823492796	4	0		0	0.7	0.7	1
1823492886	1	0		1	0.5	0.7	1
1823373596	3	1		0	0.8	0.8	0
1823493046	3	1		0	0.0	0.0	0

Included Variables

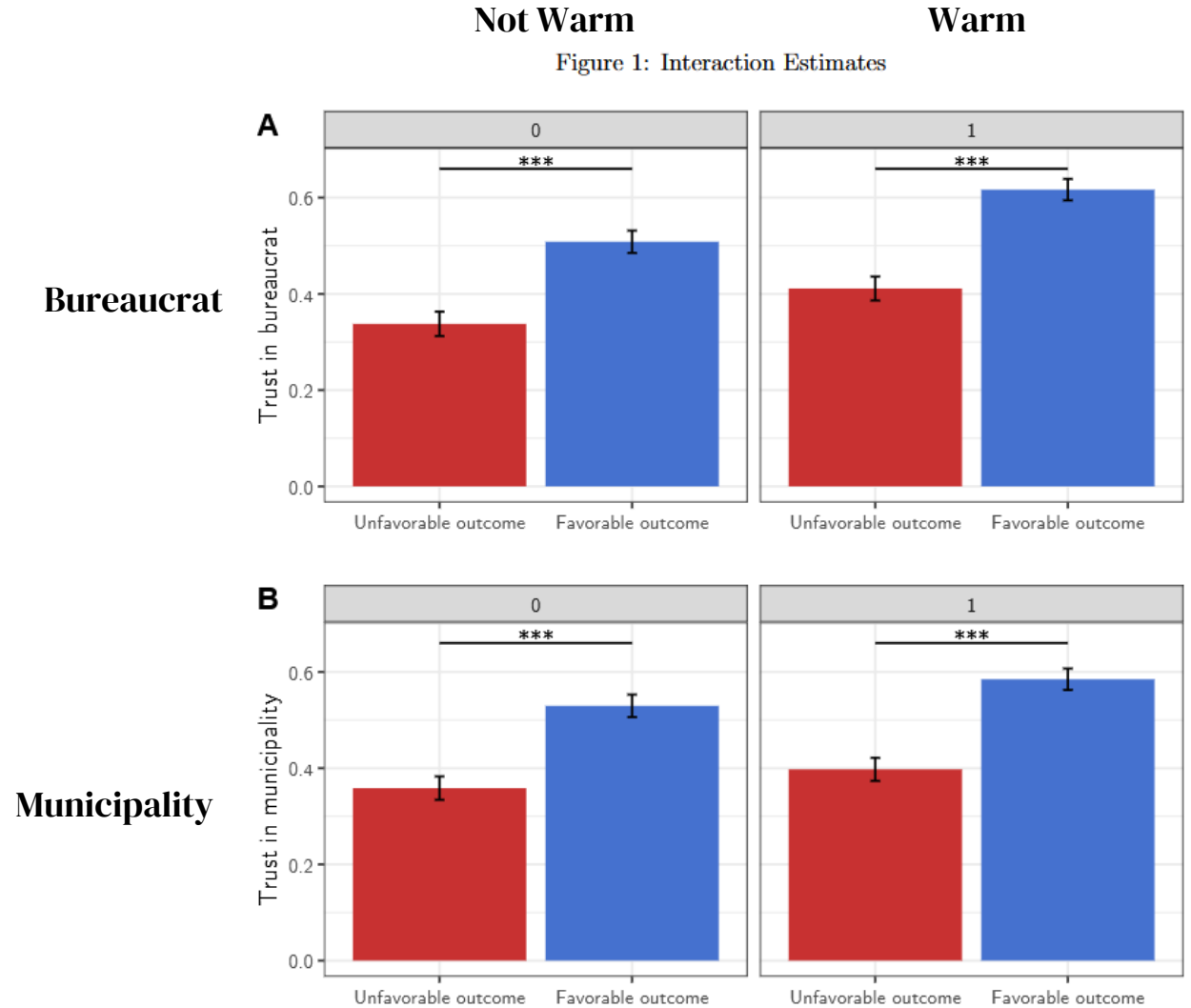
```
[1] "caseid"          "user"
[3] "pubsector_pref"  "treatment_groups"
[5] "q5"              "fair"
[7] "neutral"         "unbiased"
[9] "warm"            "friendly"
[11] "likable"         "competent"
[13] "intelligent"     "capable"
[15] "gender"          "age"
[17] "education_rc"    "ideology"
[19] "age_1"           "edu"
[21] "attention"       "trust_m"
[23] "trust_b"         "of_treatment"
[25] "warmth_treatment" "perc_warmth"
[27] "perc_fairness"   "perc_competence"
[29] "perc_favorability"
```

Overall Findings

- The level of trust in the bureaucrat and the municipality increases based both on favourable outcomes and on warmth
- Favourable outcome has a higher impact on trust than warmth in both cases
- Results for the bureaucrat and the municipality are similar

Hansen concludes that:

“public leaders should be concerned about whether their employees behave in a warm manner if they want to build greater trust”



Notes: Effect of the outcome favourability manipulation on trust in the bureaucrat (Panel A) and trust in the municipality (Panel B) conditional on the warmth treatment. Bars are means and lines are 95% CIs. *** $p < 0.001$.

OLS Results

- Positive outcome significantly increases level of trust in both bureaucrat and municipality by 18.8% points and 17.9% points respectively → **supporting H1**
- The negative effect of outcome is not reduced when bureaucrats showed high warmth versus low warmth, as interaction terms are not significant for either outcome → **not supporting H2**

Table 2: OLS Experimental Results

	<i>Dependent variable:</i>			
	Trust Bureaucrat (1)	Trust Municipality (2)	Trust Bureaucrat (3)	Trust Municipality (4)
Outcome favorability (OF)	0.188*** (0.012)	0.179*** (0.012)	0.171*** (0.017)	0.171*** (0.017)
Warmth	0.091*** (0.012)	0.047*** (0.012)	0.073*** (0.017)	0.039** (0.017)
OF * Warmth			0.035 (0.024)	0.016 (0.024)
Constant	0.329*** (0.011)	0.355*** (0.010)	0.338*** (0.012)	0.359*** (0.012)
Observations	1,516	1,525	1,516	1,525
R ²	0.163	0.136	0.164	0.137
Adjusted R ²	0.162	0.135	0.163	0.135
Residual Std. Error	0.237 (df = 1513)	0.233 (df = 1522)	0.236 (df = 1512)	0.233 (df = 1521)
F Statistic	147.665*** (df = 2; 1513)	120.190*** (df = 2; 1522)	99.197*** (df = 3; 1512)	80.248*** (df = 3; 1521)
<i>Note:</i>			*p<0.1; **p<0.05; ***p<0.01	

simple model

including interaction term

Extension - Ordered Multinomial Regression

Method

- OLS assumes the outcome variable is continuous and unbounded but the 0-10 scale is ordinal and bounded
- Four ordered multinomial regression models were run, treating the original outcomes levels 0-10 as separate categories on an ordinal scale
- Regression coefficients are the change in the log-odds of being in a higher category in each treatment

Table 5: Ordered Multinomial Regression Coefficients

	Trust Bureaucrat	Trust Municipality	Trust Bureaucrat	Trust Municipality
of_treatment1	1.409 (0.096)***	1.350 (0.095)***	1.265 (0.131)***	1.304 (0.131)***
warmth_treatment1	0.692 (0.091)***	0.333 (0.090)***	0.545 (0.129)***	0.288 (0.127)*
of:warmth			0.291 (0.181)	0.091 (0.179)

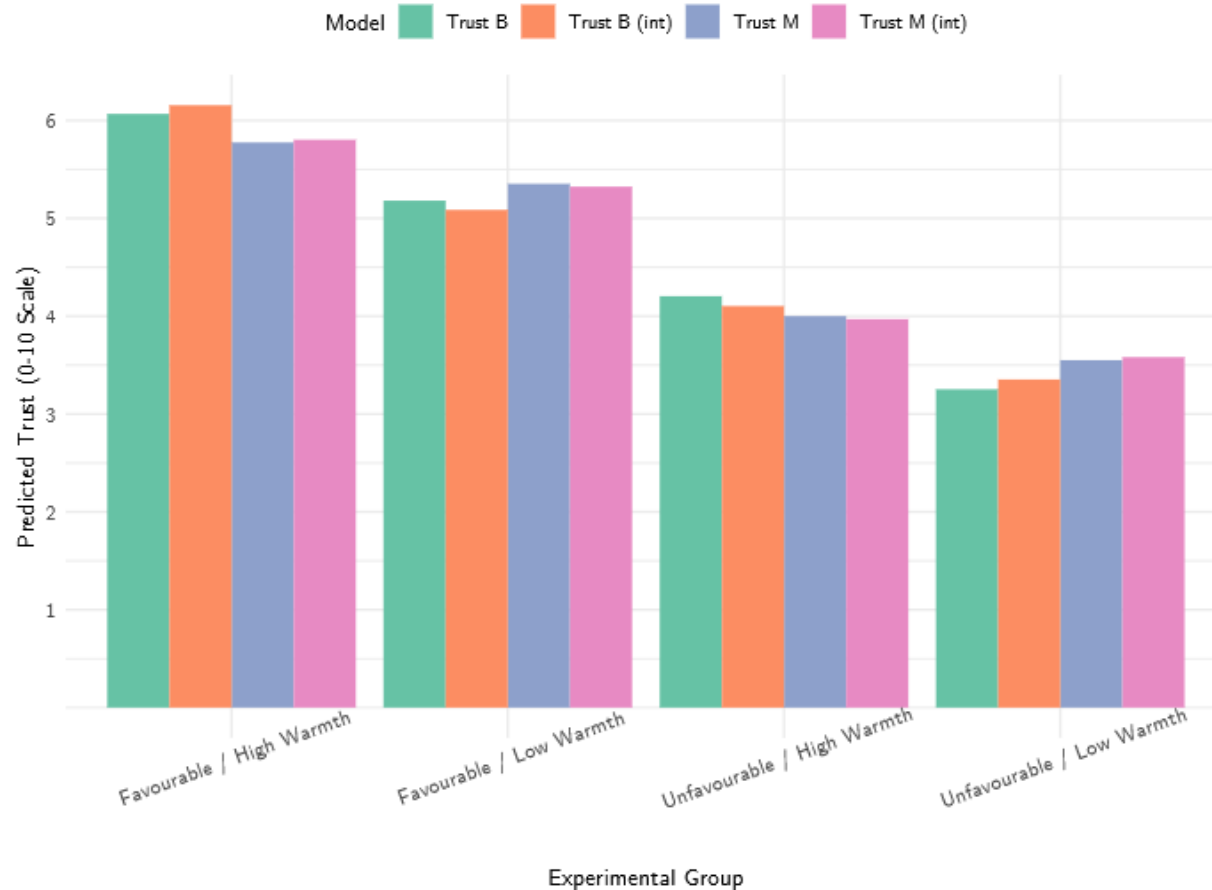
*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Outcome

- The positive impact of the treatments and insignificant interaction term are consistent with the OLS models

Extension - Ordered Multinomial Regression

Figure 2: Ordered Multinomial Regressions: Predicted Trust



- Predicted trust levels for each experimental group were calculated using the models predicted probabilities
- The predicted trust values were similar to the experimental group means
- This approach validates OLS findings and makes predictions in original scale

Predicted Group Trust Average

Group		Trust B	Trust M
1	Unfavourable / Low Warmth	3.25	3.55
2	Favourable / Low Warmth	5.18	5.35
3	Unfavourable / High Warmth	4.20	4.00
4	Favourable / High Warmth	6.07	5.78

Experimental Group Trust Average

Experimental Group	Trust b	Trust m
Unfavourable / Low Warmth	0.338	0.359
Unfavourable / High Warmth	0.411	0.398
Favourable / Low Warmth	0.508	0.530
Favourable / High Warmth	0.617	0.585

Extension - Ordered Beta Regression

```
5 library(cmdstanr)
6 library(brms)
7 library(ordbetareg)
8 cmdstanr::cmdstan_path()
9 set.seed(123)
10
11 reg_beta_b <- ordbetareg::ordbetareg(trust_b ~ of_treatment + warmth_treatment
12   , data = data)
13 reg_beta_m <- ordbetareg::ordbetareg(trust_m ~ of_treatment + warmth_treatment
14   , data = data)
15 int_beta_b <- ordbetareg::ordbetareg(trust_b ~ of_treatment * warmth_treatment
16   , data = data)
```

- Ordered Beta Regression is an alternative type of regression explicitly designed for outcome data that has an ordinal and bounded structure.
- It combines beta regression defined for any bounded continuous scale and the ordered logit that uses discrete categories.
- It is beneficial as it produces intelligible estimates that respect the bounds of the outcome variable
- The package '*ordbetareg*' estimates a Bayesian implementation finding the posterior distribution (using a Markov chain Monte Carlo)
- The predicted mean trust values for the groups were calculated from posterior predictions and show very similar prediction to the real means

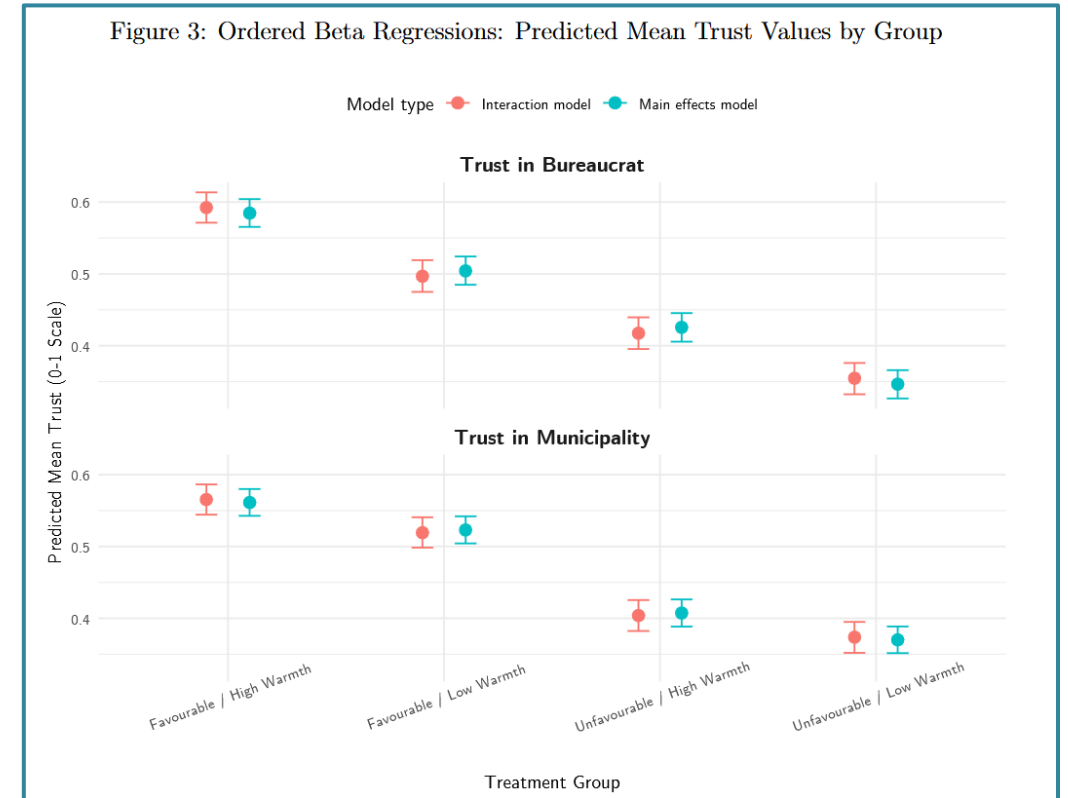


Table 8: Ordered Beta Regression: Predicted Trust of Experimental Groups with Confidence intervals

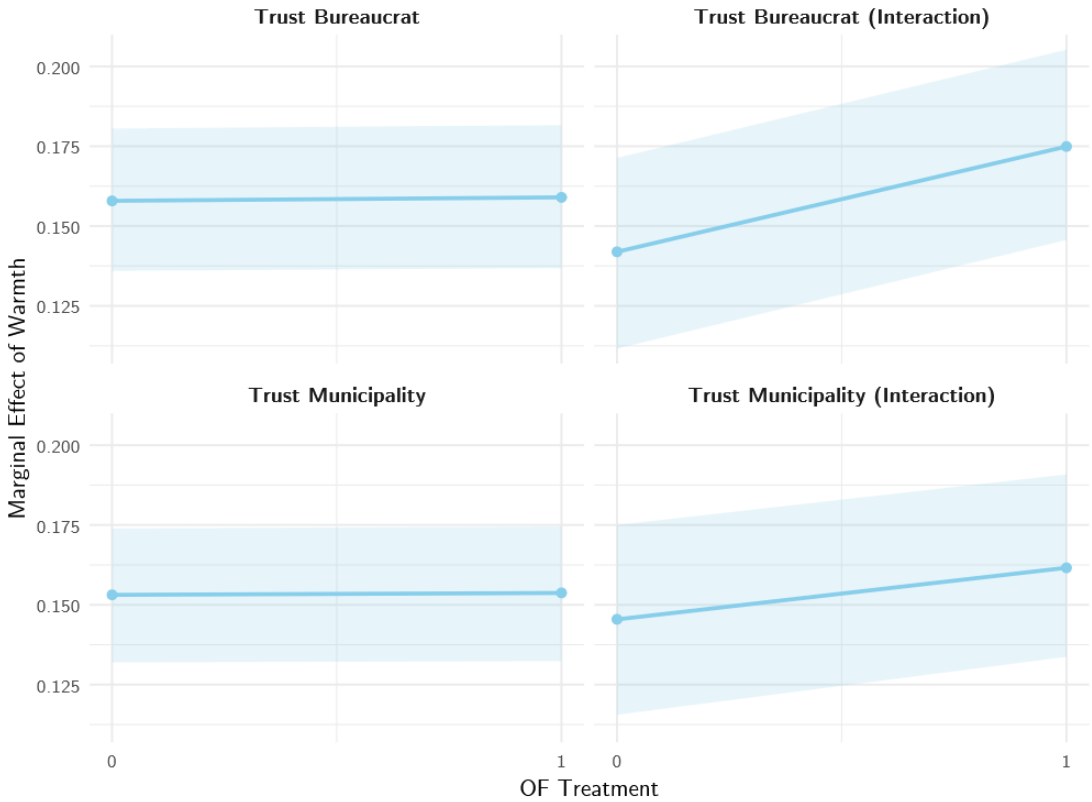
treatment	Trust Bureaucrat	Trust Municipality	Trust Bureaucrat (int)	Trust Municipality (int)
Unfavourable / Low Warmth	0.35 [0.33, 0.37]	0.37 [0.35, 0.39]	0.35 [0.33, 0.38]	0.37 [0.35, 0.40]
Unfavourable / High Warmth	0.43 [0.41, 0.45]	0.41 [0.39, 0.43]	0.42 [0.40, 0.44]	0.40 [0.38, 0.43]
Favourable / Low Warmth	0.50 [0.48, 0.52]	0.52 [0.50, 0.54]	0.50 [0.47, 0.52]	0.52 [0.50, 0.54]
Favourable / High Warmth	0.58 [0.57, 0.60]	0.56 [0.54, 0.58]	0.59 [0.57, 0.61]	0.57 [0.54, 0.59]

Conditional Marginal Effects Using Beta Regression

- The marginal effect of Warmth was calculated to understand the difference in expected trust when the warmth treatment is applied, holding the outcome favourability constant
- The results from the ordered beta predicted trust values (0-1) suggest effect of warmth increases trust in the bureaucrat and the municipality by 0.16 and 0.15 points on average.
- The effect of warmth is slightly stronger in a favourable outcome scenario indicating that warmth may be more effective in positive contexts

Table 9: Marginal Effects of Warmth Treatment by OF Treatment and Model				
of_treatment	Trust Bureaucrat	Trust Municipality	Trust Bureaucrat (Interaction)	Trust Municipality (Interaction)
0	0.16 [0.14, 0.18]	0.15 [0.13, 0.17]	0.14 [0.11, 0.17]	0.15 [0.12, 0.17]
1	0.16 [0.14, 0.18]	0.15 [0.13, 0.17]	0.17 [0.15, 0.21]	0.16 [0.13, 0.19]

Figure 4: Ordered Beta Regressions: Marginal Effects of Warmth



Conclusion

- The original study was replicated successfully confirming the results that outcome favourability and warmth strongly influence citizens' trust in government.
- Two alternative methods were applied: an ordered multinomial and an ordered beta regression
- The ordered multinomial reflected the ordinal nature of the data and found similar treatment effects to the OLS model, made accurate predicted values and further supported the original findings that there is not a significant interaction effect
- The ordered beta regression using Bayesian estimation reaffirmed the previous findings but with an approach that is specifically designed for bounded ordinal nature of the outcome variable

Thank you for listening!

Any Questions?

